Subject: Programming with Java

ASSIGNMENT No : 1

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BATCH:CSE B3
ACADEMIC YEAR: 2025-29

TITLE :

Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

AIM:

To develop a Java application using the Java environment setup and demonstrate the concepts of JDK, JRE, JVM, variables, methods, and constructors. The program also aims to understand the Java execution process and implement these concepts through suitable examples.

PROGRAM CODE:

```
public class Main {


    static int square(int number) {
        return number * number;
    }

    public static void main(String[] args) {


        String name = "Suhani";
        int age = 19;
        double salary = 550000;


        int result = square(6);


        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("Salary: " + salary);
        System.out.println("Square of 6 = " + result);
    }
}
```

OUTPUT:

```
Last login: Sun Jul  5 14:45:14 on ttys029
[suhanigarg670@Suhanis-MacBook-Air ~ % cd Desktop
[suhanigarg670@Suhanis-MacBook-Air Desktop % javac Main.java
[suhanigarg670@Suhanis-MacBook-Air Desktop % java Main
Name: Suhani
Age: 19
Salary: 550000.0
Square of 6 = 36
suhanigarg670@Suhanis-MacBook-Air Desktop %
```

QUESTIONS:

Q1: Explain the main concept used in Java environment setup, variables and methods.

JDK (Java Development Kit) is used to develop Java programs. It contains the compiler (`javac`), JRE, and other development tools.

JRE (Java Runtime Environment) provides the environment required to run Java programs. It contains the JVM and Java libraries.

JVM (Java Virtual Machine) executes the Java bytecode by converting it into machine code, making Java platform independent.

The program also demonstrates the use of variables, which store different types of data such as integers, strings, and decimal values.

A method is a reusable block of code that performs a specific task. In this program, the `square()` method calculates the square of a number and returns the result.

The experiment also introduces constructors, which are special members of a class used to initialize objects automatically when they are created.

Q2: What are the advantages of using Java variables and methods?

Advantages of Variables

- Store different types of data efficiently.
- Make programs easy to read and understand.
- Allow values to be modified easily.

- Reduce repetition of values in the program.

Advantages of Methods

- Promote code reusability.
- Reduce code duplication.
- Improve program readability.
- Make programs modular and organized.
- Simplify program maintenance and debugging.

Q3. How can this implementation be improved?

1. Accept user input using the Scanner class instead of fixed values.
2. Use constructors to initialize object data.
3. Add more methods to perform additional operations.
4. Use exception handling to handle invalid input.
5. Add proper comments for better understanding.
6. Use meaningful variable and method names.
7. Organize the program into separate classes following OOP principles.

Q4.What are the real-world applications?

Student Management System – Stores student details and marks.
Banking System – Manages customer accounts and transactions.
Hospital Management System – Maintains patient records and appointments.
Employee Management System – Stores employee information and salary details.
E-commerce Applications – Manages products, orders, and payments.

CONCLUSION:

The experiment was successfully completed and demonstrated the Java environment setup along with the use of variables, methods, and constructors. It improved the understanding of Java program execution and basic programming concepts.